

**ESTIMATING THE ECONOMIC VALUE OF WOMEN'S UNPAID
DOMESTIC AND CARE WORK IN INDIA: EVIDENCE FROM
TIME-USE SURVEY DATA AND ITS IMPLICATIONS FOR GDP
MEASUREMENT**

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1. Introduction

1.1 Background of the Study

Unpaid domestic/care work is an essential but systematically under-estimated part of economic activity. It encompasses activities associated with cooking, cleaning, child care, elder care and other household activities that support day-to-day life activities and contribute to the operation of the labour market. Although it is central, this work is mostly not visible in the traditional economics accounting system, especially the Gross Domestic Product (GDP), which only acknowledges transactions of a market nature. The implication of this exclusion is immense in the understanding and measuring of economic productivity and contributions within society.

In the Indian context, the division of labour continues to be gendered immensely since there is a greater likelihood that women may be engaged in unpaid labour. The recent data of the Time Use Survey (2019) encompassed over 299 minutes on the time-consuming housework of the Indian female workers per day versus the 97 minutes on the time-consuming housework of male workers, which is a substantial depiction of gender inequality. On the same note, the unpaid caregiving roles of women consume a significant amount of time from the traditional gender roles at home. Such perception deterred women from pursuing official jobs and reduced their representation in the official data of the economy. In addition, participation of women in the Indian labour force is also quite low and even in recent years, the level of participation registered was 25-30 per cent in the level of participation which has also been partially contributed by the restriction of the unpaid tasks.

In a more macro-economic sense, unpaid domestic and care work is very important in maintaining the human capital and facilitating the productive economy. It facilitates reproduction of the labour force as it guarantees the welfare of the household members, their health and education. Nevertheless, its omission in GDP contributes to the understatement of the overall economic performance and policy making towards gender biases (Samantroy., 2022). Feminist economists have also suggested that the omission is symptomatic of a structural weakness of conventional economic frameworks, which focus on market transactions rather than on non-market contributions.

Of very critical concern, the inability to incorporate unpaid labor will not only misrepresent economic indicators but will also affect policy priorities, overshadowing investments in care

infrastructure and social support systems. With the rising prominence of inclusive growth and gender equality debates, the growing acceptance of the need to include unpaid labor in national accounting systems is developing, in the form of satellite accounts or other forms of valuation. This is needed in order to have a more complete and fairer picture of economic activity in India.

1.2 Problem Statement

The mainstream measure of the economy's performance, Gross Domestic Product (GDP), systematically fails to represent unpaid domestic and care work in the national income accounting. This exclusion leads to a gross underestimation of the economic activity, especially that by women. In India, a recent Time Use Survey (2019) shows that on average, a woman spends more than 299 minutes a day on unpaid household services and caregiving, compared to 97 minutes by a man, a significant gender gap in time allocation. Although this kind of labour is vast and is needed to maintain homes and sustain the formal economy, it is still economically invisible.

It has been estimated worldwide that an all-encompassing analysis could see some countries as developing, especially India, at the upper end of the scale, with 9-15% of GDP potentially attributed to care work which was not remunerated accordingly. Such work is not counted in GDP, and not only are the macroeconomic indicators distorted, but the undervaluation of the productive efforts of women by the labor market also endorses structural gender inequalities. Moreover, this weakness restricts evidence-based policymaking, in that it does not reflect the extent of labour involvement and economic well-being. Thus, the continued failure to acknowledge unpaid work is a conceptual and methodological gap in the frameworks of national accounting and requires serious re-examination.

1.3 Research Objectives

The Time-Use Survey data would form part of the study design, whereby the level of poverty of women involved in the unpaid household and care work in India would be based on the Time-Use Survey data, on which approximately 85-90 per cent of all unpaid hours of care work is estimated based on the current national data of such on the same. It also falls into the socio-economic determinants of distribution and value of unpaid labour, i.e. gender, age, residence, employment, education and marital status. The implication of inclusion is likewise critically analyzed regarding national income theories of evaluation of unpaid work on the premise that

inclusion of unpaid work can drastically affect the habitual gendered GDP indicator, and economic invisibility.

1.4 Research Questions

This paper is informed by three research questions, which are interrelated, that critically analyze the economic invisibility of unpaid work.

- What is the approximate economic cost of the domestic and care work that women in India do unpaid, considering that women spend about 4.5 to 5 hours a day on such activities, relative to less than 1 hour that men spend on them, based on recent time-use data?
- What socio-economic variables, including gender, age, education, employment status, and urban-rural residence, play a crucial role (and determine) the scale of this unpaid contribution?
- How would the integration of this value, which may explain 15-40 per cent of GDP, change standard indicators of national income and economic productivity?

1.5 Significance of the Study

The second implication of the importance of valuing unpaid domestic and care work by women is the policy relevance of the results, as demonstrated by recent Time Use Survey (2019) data that showed that women spent an estimated 299 minutes a day on unpaid work as opposed to the 97 minutes by men. Although a major contribution, such work is not counted in the national income accounting systems, which has led to systematic underestimation of the economic participation of women. The current work fits into the policy discourse by offering empirical estimates that can be used in policy formulation, gender-responsive budgeting, and labour policies, especially in a country where the participation rate of the female labour force is still below 25%, based on current estimates.

In gender equality terms, the invisibility of unpaid work is strengthening structural inequality by losing the value of the time of women, which means their restriction to a paid work opportunity (George and Shaji., 2024). Assigning a monetary value to unpaid work reveals the opportunity cost incurred by women critically and the existing economic beliefs that equate productivity to market-based work alone.

Moreover, the research has implications for how the national accounting systems can be enhanced. The modern GDP data, as computed in accordance with the System of National Accounts (SNA), does not capture non-market household production, and therefore may drop 15-40 per cent of the economic value as estimated in the more encompassing international calculation. The inclusion of such estimates in the form of satellite accounts may enhance more substantiated evidence-based policymaking by giving a better understanding of economic activity and welfare.

1.6 Structure of the Dissertation

To support a prospective and logical flow of analysis, this dissertation is divided into six chapters. The first chapter gives the problem, objectives and the importance of our research. Chapter 2 is a critical review of the extant literature, pointing out theoretical reflections as well as gaps in knowledge that need to be bridged in order to value unpaid domestic and care work. Chapter 3 details the research methodology, such as sources of data to be used, specification of variables and the econometric methods that were employed in the research. Chapter 4 includes empirical findings of the time-use Survey analysis based on Time-Use Survey data and using Stata. Chapter 5 presents a critical discussion of findings, giving them connections to the broader economic and policy debates. Lastly, Chapter 6 also presents some of the main pinpoints, limitations, and suggestions of subsequent research.

2. Literature Review

2.1 Conceptual Framework of Unpaid Work

An essential but systematically undervalued element of the economy is the unpaid domestic and care work. In traditional economic accounting systems, especially national income accounting systems, these types of activities cannot be seen as they play significant roles in the maintenance of households, labor market systems, and the overall economic productivity. The idea of unpaid work has gradually changed over the years, especially with the input of international organizations that have tried to define how close it can go to its outer limits, including international labour organizations like the ILO (International Labour Organization), and international systems of national accounts like the United Nations System of National Accounts (SNA).

Definitely, unpaid work can be seen as all the productive activities that are performed without payment (usually at home or in society) and which serve the interests of both the individual and society. The ILO conceptualises unpaid work as both unpaid domestic services and unpaid caregiving work, acknowledging that these services are necessary to reproduce the workforce. In the same vein, the SNA separates production within the general production boundary from that which should not be included in calculating GDP. Although goods consumed themselves (including subsistence farming) are captured, most of the unpaid services (including cooking, cleaning and caregiving) are not included in this, which is a structural constraint in the GDP as a total indicator of economic activity.

A recent estimate in the global scene emphasises the value of unpaid work. According to the time-use survey data, women are involved in about 75-80 per cent of the total amount of unpaid care work around the world. In the Indian case, the Time Use Survey (2019) indicates that women, on average, spend 299 minutes daily on domestic services but are not paid, as opposed to men, who spend merely 97 minutes daily (Sahu., 2024). Women in unpaid caregiving activities get about 134 minutes a day, and men get about 76 minutes in this kind of activity. The gendered nature of unpaid work and its strong interrelationship with social norms and institutional arrangements are highlighted by these inequalities.

There are two main types of unpaid work, namely domestic work and caregiving work. Routine chores related to household cleaning, cooking, washing, and maintenance are part of domestic

work. Although repetitive and, in most cases, time-consuming, these are the tasks that are necessary to ensure the continuity of households. Care-giving work, however, is direct care that is given to those who are dependent, either children, the elderly, or those who have disabilities. In comparison with domestic work, caregiving is more complex since it may involve emotional labor and round-the-clock attention; thus, it is a more complex and less replaceable type.

A critical examination of these categories shows that there are some fundamental issues with measuring and valuing them. Homework might be a more feasible indicator of what can be measured because the activity tends to be routine, but caregiving is a qualitative aspect challenging to measure using time-use data alone. Moreover, the fact that unpaid labor The fact that it does not constitute a section of GDP not only contributes to the deterioration of overall economic performance but also causes a feeling of inequality between genders, as women are not regarded in terms of their role in the economy. This drawback of usefulness raises certain valid questions of whether GDP is an adequate measure of welfare and hints that additional systems of account, such as satellite accounts, may be necessary to make economic activity totals more complete.

In brief, the conceptual framework of unpaid work includes a conflict between the economic significance, which can never be ignored and the marginalisation of formal measurement systems. Although international guidelines have gone far in defining and classifying unpaid work, there is still much room to integrate it into mainstream economic analysis, and a more comprehensive and gender-sensitive approach to national accounting is necessary.

2.2 Theoretical Perspectives

The problem of economic analysis of unpaid domestic and care labour was addressed in the framework of various theoretical schools, and they introduced their own vision of how the process of conceptualization and quantification of unpaid domestic and care labour could occur. The most critical ones are household production theory and time allocation theory. This group of frames gives a minimal understanding of why such a major problem as unpaid work, although so important to society in such a fundamental manner, is never systematically underestimated in the traditional economic accounting.

The traditional scope of economic analysis is disrupted by feminist economics because it seeks the inclusion of unpaid labour in the national income calculation. It contends that all of the conventional economic models, especially the models that form the basis of the Gross Domestic Product (GDP), have a structural bias of market-driven operations and do not credit the economic value of non-market labor, the majority of which is performed by women. Empirical data indicate that the unpaid home and care services would represent a significant percentage of the economic output if they were monetized. Indicatively, time-use data available in third-world countries frequently show that women devote 3-10 times the number of hours of unpaid labour that men spend working at home on a daily basis (Nikore., 2022). Critically, the feminist economists believe that such invisibility causes distortion in the policies that strengthen gender inequalities in the labour markets and the social protection frameworks. Nonetheless, as much as feminist economics has managed to point out the normative and structural biases, it has had its share of criticism due to its lack of a mechanism for providing broad agreement on standardised methods of measurement, thus making it difficult to incorporate into conventional economic measurement systems.

Household production theory, presented by Becker (1965), is an elaboration of neoclassical economics in the sense that it theorized households as production units that utilize time and market resources to produce commodities, including meals, childcare, and household repairs. This structure offers a more formal way of including unpaid work within economic analysis by considering it a productive activity, and has the effect of creating utility. In this model, people will spend their time on market and non-market activities according to the relative wages and preferences. Although this method can be used to implicitly value unpaid work (typically using opportunity cost or replacement cost techniques), it is also critiqued on the assumption of rational decision-making and efficient allocation at the household level. In practice, gender structures and power relations often limit the options of women and consequently result in unbalanced, unpaid labour, which is not completely represented in the model. Moreover, the fact that household production can be directly equivalent to market production does not consider the qualitative disparities, especially in the sphere of care work, where emotions and relationships play an important role.

The theory of time allocation is based on this platform, but it pays specific attention to the way individuals allocate the scarcity of their time to competing interests with time spent in paid employment, unpaid employment and recreation. Since time is a scarce resource: 24hours a

day, this methodology will offer a quantitative methodology of trade-offs in labour supply choices. Surveys on time-use, like surveys in India, indicate that women usually have to reduce 4 to 6 hours daily (40 to 6 hours) to unpaid domestic housework and care, compared to under 1 to 2 hours daily (less than one-third of the day) by men. This inequality highlights quantitative limitations of women within the formal labour market. The theory of time allocation is especially helpful in empirical studies because it enables direct measurement of unpaid work in time inputs; it is seen as recording this unpaid work. Nevertheless, it is more inclined to measure time rather than to qualify the work, and it may underestimate the acuity and multidimensional character of caregiving tasks. Also, it does not necessarily deal with the social and institutional aspects, which determine time-use patterns.

Synthetically, even though feminist economics offers a critical normative framework that emphasizes systemic undervaluation, the hypothesis of household production is a formal economic model of the need to bring unpaid labour into analysis, and time allocation is a theory that provides the methodological instrumentation required to measure unpaid labour (Chakraborty., 2022). Both frameworks offer great information, but also demonstrate certain limitations, which implies that a hybrid solution is crucial to understanding unpaid domestic and care work better and the impact on economic measurement.

2.3 Measurement Approaches to Valuing Unpaid Domestic and Care Work

The most significant research methodological quandary in economic studies has perhaps been the unvalued household and care labour since this type of economy has not been part of the reporting system in the mainstream economies. Literature has identified three important techniques that have been developed over the years as the opportunity cost technique, replacement cost technique and the output-based valuation technique. The variations between the two approaches are found in the conceptual origin, the restrictions, and the implications, which are empirical.

Opportunity Cost Method

According to Roy and Sekher., 2025, opportunity cost approach quantifies unpaid work in terms of the revenues that an individual foregoes when participating in opportunities beyond the market, rather than pursuing paid employment. In real life, this is pegged on the level of a wage rate on what the individual would have earned in the labour market. An example is when

an educated woman might be earning, say, 300-800 rupees per day in the formal or informal sector (according to the current trend in Indian wages by occupation), her unpaid housework would be worth one such level of income.

This methodology is based on the neoclassical theory of labour supply, which supposes that people rationally spend their time in market and non-market activities. Although this method has great conceptual appeal, there are a number of limitations to it. First, it is likely to overvalue the worth of the unpaid work of highly skilled people because their potential income can be much higher than the market prices of domestic services (Roy and Sekher., 2024). On the other hand, it can underprice labour done by members of the labour market who have been less favoured by different factors and especially rural females, as their observed wages are not low as a result of ineffectiveness; they are low as a result of structural disparities.

Furthermore, it presumes that employment opportunities are realistic and equally accessible to all people, and this is usually not applicable in developing economies like India. Participation of the female labour force in India is still quite low (in the latest statistics, it is about 25-30 per cent), which means that not all women have a good working option. Consequently, the opportunity cost technique can capture hypothetical as opposed to real trade-offs in an economy, and this weakens its empirical validity.

Replacement Cost Method

The replacement cost technique determines the worth of the uncompensated work as the price of outsourcing a person to do the same job in the market. One of the most common strategies is this one because it is based on the measurement in practice and can be matched to the observable market wages. The wage rates of domestic workers in the Indian scenario are usually between 5000 and 12 000 per month, as per region, degree of urbanization, and service (cooking, cleaning, and childcare). This equates to around 200-500 a day in the case of basic domestic services.

This strategy is of two types: a generalist strategy (the wage of a domestic helper is used), and a specialist strategy (the wages of various types of work (e.g. childcare worker, cook, cleaner) are paid at different rates). The specialist approach is more accurate but involves more time-consuming data and wage data.

In empirical studies, the replacement cost method is frequently used in preference over the factor endowment method due to its reliance on the actual cost of substituting unpaid labour in the actual market, which makes it more realistic about the actual economic transactions. Nevertheless, it does not have unlimited possibilities (Sahoo *et al.*, 2024). It can underestimate non-replaceable in the market complex care activities like emotional support or elder care. Also, household work commonly entails multitasking, efficiency benefits that cannot be easily replaced by hiring labour, which might be underestimated.

These limitations notwithstanding, this is a method that is usually regarded as the most policy-relevant, especially when estimating the contribution of unpaid work to the extended or satellite GDP accounts.

Output-Based Valuation Method

The output-based one determines the value of unpaid labour by deducting the actual value of actual output in actual production of goods and/or services and prices in the market. Natural cooked meals can then be matched with home-equivalent market and childcare-equivalent prices of similar food in the market and childcare, estimated respectively towards the price of daycare services.

Theoretically, the approach is very similar to the production boundary that is established in national accounts in the sense that it emphasizes on quantifiable outputs as opposed to inputs (time). There is, however, a great limitation in data usage for its application. Proper valuation needs precise statistics about the amount of outputs and their quality, which is seldom known in a survey of such magnitude as the Time Use Survey in India (2019).

Secondly, household production is usually joint production and non-marketed, and therefore it is hard to isolate and price-segment production. Care giving is also an example, which does not involve physical and emotional components that are easily measured. This has also contributed to the output-based approach failing to gain popularity in the empirical studies, particularly in the context of developing countries.

Comparative Evaluation

Through the critical comparison of the three approaches, it can be seen that no one approach gives a complete, comprehensive valuation. The opportunity cost approach is conceptually

sound but limited by fees in the imperfection of the labour market. The replacement cost approach is useful in its application, yet it could also be inaccurate in the assessment of the qualitative aspects of care work. The output-based measure is aligned with national accounting, but is seriously data-starved.

Considering these trade-offs, the replacement cost method serves as the main method of valuation in most modern research, especially in the Indian context. It is the most plausible and policy-related approach to scale the economic worth of unpaid domestic and care work due to its dependence on actual market wages and its applicability with time-use information.

2.4 Time-Use Surveys in India

Time Use Survey (TUS) 2019, carried out by the National Statistical Office (NSO), comprises the most detailed and up-to-date information on how time has been distributed among different activities in India. It is estimated that around 1.4 lakh persons 6 years and over years of rural and urban areas, giving nationally representative estimates regarding the duration spent on paid and unpaid activities. The survey is especially valuable as it allows measuring the size of non-paid household and care labour, which have been mostly undetected in traditional economic accounting models, including GDP.

According to the findings of the TUS 2019, gender inequality in the allocation of non-waged work is high. This domestic service, not paid among the Indians, puts more burden of the decision upon the women of India since they spend more domestic time on an average of 299 minutes a day compared to 97 minutes spent by the Indian men (Majumder and Som., 2025). The time spent on unpaid care/caregiving services is nearly an entire day (134 minutes daily) by women and 76 minutes by men. These differences illustrate that women are nearly 3 times more likely to engage in unpaid home labour in comparison with men and therefore that there is a gendered division of labour.

Men, on the other hand, spend much more time at work and in associated affairs. According to the survey, men work an average of 263 minutes a day in paid employment, whereas women work only about 92, showing gaps in labour force engagement and access to market-based employment. This disparity indicates that the unequal contribution of women to the unpaid labour would limit their participation in the formal economic activity of the economy, prejudicing structural inequalities.

Critical analysis of these results suggests that the TUS can be useful to provide quantitative data, but it might still underestimate the actual burden of unpaid work because of possible problems like underreporting, social desirability, and underreporting of multitasking activities. Moreover, the survey also fails to directly attach a monetary value to unpaid work, and this limits its use in, as it is, to inclusion in national accounts. Irrespective of these weaknesses, the TUS 2019 can be an important empirical basis of any study that seeks to determine the approximate economic value of unpaid labour and find its practical implications on GDP measurement.

Generally, the evidence confirms the systematic underestimation of women's contributions to the economy and drives the point to include unpaid work in the wide scope of measures of economic performance.

2.5 Empirical Studies

India-based Studies

Empirical work estimating the economic value of in-home and care work without pay in India has grown dramatically since the publication of the Time Use Survey (TUS) 2019. This database, which is nationally representative, gives fine-grained information on the allocation of time by people in paid and unpaid forms to make it easier to quantify the contribution of women in the non-formal economy.

It has been demonstrated that there is a strong gender difference in terms of time allocation. This period is about 299 minutes a day, according to the TUS 2019 data, spent by women in India on domestic services, and about 97 minutes in the same age group spent by men. Moreover, women spend approximately 134 minutes a day in unpaid caregiving, compared to 76 minutes by males. Such statistics underscore the unequal weight of household labour assigned to women, which supports old structural patterns of unequal household labour allocation.

Some empirical studies based on this data have tried to monetize unpaid work, typically based on the replacement cost measure, usually using the current wage rates, which is used in domestic work. It is estimated that the economic price of unpaid work of women in India varies between 15 percent to 40 per cent of GDP, depending on the method of valuation and

assumptions about wages (Goldar *et al.*, 2025). Conservative estimates generally estimate this as 2025% of GDP, which is large compared to other large formal careers like manufacturing or agriculture.

Critical analysis of these studies, however, shows that there are inconsistencies in methodology. To begin with, the wage rate will have a great impact on valuation. Surveys that use minimum wage rates come up with lower estimates, and those that use specific service wages (e.g., childcare or eldercare professional) give higher estimates. This brings the issues of comparability and standardization across studies.

Second, a lot of Indian studies are of the aggregate type, which are usually not based on heterogeneity among the socio-economic groups. As an example, rural women are likely to find themselves doing more unpaid labor than their urban counterparts because they lack access to labour-saving technologies and communal amenities. The same can be said of education and employment status, which determine patterns of time allocation, but are not necessarily strictly included in empirical models.

Third, though other studies are trying to do regression analysis to determine determinants of unpaid work, there is still no integration of survey-level results and macroeconomic consequences. In particular, not many studies systematically relate individual-level estimates of the unpaid labor with the frameworks of national income accounting. Consequently, the policy relevance of such findings has not been well developed.

Irrespective of such shortcomings, Indian empirical studies have been such that it has initiated the invisibility of the work of women in traditional economic measurement. It has additionally led to policy debates on gender-sensitive budgeting and whether unpaid work should be included in satellite accounts or not.

Global Comparisons

Empirical research on unpaid domestic and care work follows the same trends all over the world, albeit with some differences in the level and methodology used. Similar frameworks are created by time-use surveys carried out in different countries to address the issue of gender inequalities in unpaid work.

Women in developed economies on unpaid work spend less time on unpaid work than women in high-income countries, but the gender gap still exists. To illustrate this point, the average amount of time spent by women on unpaid work in OECD countries is about 3-4 hours per day, whereas males spend about 1.5-2.5 hours. Conversely, the amount of labor women do without pay in developing countries, South Asia and Sub-Saharan Africa, can be more than 5 to 6 hours a day, which is concerning the disparities in the infrastructure, social norms and labor market.

The global level of valuation studies suggests that unpaid labor produces 10 per cent to 60 per cent of GDP, based on the country and methodology applied. As an example, estimates of most countries, such as Australia and Canada, are usually between 30 and 50 per cent of GDP, whereas lower estimates are found in those with less complete data coverage. Such broad ranges of variability emphasize the sensitivity of the outcome to methodological decisions, especially the valuation method used.

An important point of difference in the global research is the use of satellite accounts that are increasingly employed in order to include unpaid work in national accounts. Some countries, like the United Kingdom and Australia, have put up experimental frameworks to estimate household production, offering a more systematized means to incorporate unpaid labor into the economic analysis (Hanna *et al.*, 2023). This is opposed to India, where such an integration is still in a conceptual stage and is not yet institutionalized.

In addition, several more sophisticated econometric methods are being employed in global studies to determine how to examine determinants of unpaid work, such as panel data models and time-use regressions. The approaches enable a more subtle discussion of how variables like income, education, and labour market membership have an impact on time distribution. Comparatively, numerous Indian studies continue to be based on much akin cross-sectional frameworks, constraining the level of analysis.

The other significant difference is the policy implementation of the results of empirical findings. In several developed nations, policies surrounding parental leave, childcare support and work-life balance have been informed by the evidence of unpaid work. However, unlike in India, although the nation has acknowledged the role played by unpaid work in policy formulation, its empirical evidence has not been translated into elaborate policy interventions yet.

The point is that the task of equitably reflecting on the extent of unpaid work is also noted in global literature. Emotional labor and supervisory care activities are seldom reported in time-use surveys, thus being underestimated in overall contributions. Such a restriction can be found both in developed and developing settings, which is why better data collection methodologies might be required.

2.6 Research Gap

There are considerable gaps in the empirical and methodological literature in the context of unpaid domestic and care work, even though the discussion of this type of work is gaining more and more acceptance in the development field. The available literature has mainly been based on aggregate-level estimates, based on national accounts or descriptive statistics based on time-use surveys. As an example, the Time Use Survey in India (2019) shows that women dedicate about 299 minutes to unpaid domestic services every day in comparison to males who make 97 minutes (which reveals profound gender differences). Nonetheless, the bulk of the literature is able to record the differences without trying to genuinely analyze the socio-economic determinants at the micro level.

In addition, although the methods of valuation (especially the replacement cost method) have estimated that unpaid work may contribute up to 15 to 39 per cent of GDP in developing economies, there is little incorporation of these valuations into econometric models (Das and Nayak., 2022). Namely, the absence of regression-based studies that measure the systematic impact of the individual characteristics in terms of their gender, education, employment status, and location has been uncovered.

The other severe gap is the ineffective connection between the micro-level valuation and macroeconomic indicators. Policymakers often discuss the omission of unpaid labor in their GDPs, but very few studies have tried to empirically model its inclusion or what it would mean to the national income accounting. This lack of relevance limits the applicability of the available research regarding policy, and the importance of combined methods of analysis that can help connect household-level data with macroeconomic measurement platforms should be highlighted.

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